# The Z File System: It's Honest and Different

The Z File System, or ZFS, is an advanced file system designed to overcome many of the major problems found in previous designs. Initially developed by Sun Microsystems, its development has now been moved to the OpenZFS Project.

Data is priceless. Small ventures to the fortune 500 spend crores to maintain backups and ensure data integrity. The loss of even a byte of data can impact huge MNCs and could possibly threaten a nation's security. Ever since the evolution of computers and file systems, experiments and research have been going on in an effort to find an efficient, trustworthy backup system.

## What is the Zettabyte File System (ZFS)?

ZFS is an advanced, open source file system and logical volume manager designed by Sun Microsystems for use in its Solaris operating system, and licensed under the Common Development and Distribution License (CDDL). The name ZFS is a registered trademark of the Oracle Corporation.

## Why should you use ZFS?

The answer to this question lies in the three main design goals, which have translated into ZFS'most appealing features.

**Data integrity:** Integrity is the keyword that any serious user searches for. In ZFS, all data includes a checksum of the data. When data is written, the checksum is calculated and written along with it. When that data is later read back, the checksum is calculated again. If the checksums do not match, a data error has been detected. ZFS will attempt to automatically correct errors when data redundancy is available.

**Pooled storage:** Physical storage devices are added to a pool, and storage space is allocated from that shared pool. Space is available to all file systems, and can be increased by adding new storage devices to the pool.

**Performance and scalability:** It is a 128-bit file system that's capable of managing zettabytes (one billion terabytes) of data. Multiple caching mechanisms provide increased performance. Examples include ARC, an advanced memory based read cache; L2ARC, a second level disk based cache; and disk-based synchronous write cache, ZIL.

In addition to these, ZFS supports multiple RAID levels (redundant array of independent disks) that further improves performance and diminishes redundancy.

## Installing ZFS in Ubuntu

In Ubuntu, ZFS can be installed by running the following code:

```
$ sudo add-apt-repository ppa:zfs-native/stable
```